Department of Computer Science and Engineering
MIDTERM EXAMINATION
SUMMER 2023
CSE 340: Computer Architecture

Total Marks: 25 Time Allowed: ?? Hours

- **Answer ALL the questions**
- **At the end of the exam return both your answer script along with your question**

Name: _____ ID: _____ Section: _____

1.	CO1	a.	Define a multiprocessing system with diagram		**3**
		b.	Explain the power trend equation.		**3**
		c.	i.	If an operation needs 150 seconds in total and among that 15% is serial operation. If you want to improve the performance by 3.75 times, what improvement you need to include in your system.	**3+2**
			ii.	Draw the diagram of the von Neumann architecture.	
2.	CO2	a.	In 32-bit MIPS architecture how many R-type instructions are possible.		**2**
		b.	Design a register file having 256 registers and each register can hold 128-bit of data. Make sure that your bit lengths in each input and output links show the correction configuration.		**3**
		c.	If base address of an array A is 0x00001234, then calculate the starting address of array A [24].		**2**
		d.	Write MIPS code from the following code sequence. Consider base address of A is in $s0 and X and Y are in registers $s1 and S2 respectively. Make sure you make optimum utilizations of the registers. A [7] =13*X+4*A [5] +11*Y-100;		**3**
		e.	Consider the conditional branch instruction **Bne $10, $11, 100**. Assume program counter value is 0x00002222.Calculate the branch target address. Your answer should show all the steps in detain will necessary diagram.		**4**